BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

LESSON 185 · Insertion sort: trace and implement

Name: _____ Class: _____ Date: _____

Time: _____ Teacher: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. What is [5,2,4] after one complete ascending bubble-sort pass?

Q2. What is the value of i after i++ when i was 2?

Q3. Is a stack FIFO or LIFO?

Q4. What change makes bubble sort terminate early on sorted data?

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

I Do · Read new knowledge

Key vocabulary

key — Value being inserted into the sorted prefix.

sorted prefix — Initial segment already placed in order.

shift — Move a value one index to create space.

in-place — Uses little additional data storage.

stable — Equal elements can preserve relative order.

New learning

One key at a time — The prefix to the left of i is already sorted; take a[i] as the key.

Make room — Shift larger values in the sorted prefix one position to the right.

Insert the key — Place it at j+1; the sorted prefix grows by one element.

Worked example

Insert key 1 into the sorted prefix [4,8] of [4,8,1,6].

1. Predict first — Identify the shifts needed.

2. Model solution — Shift 8 then 4 to the right and insert 1 at index 0 → [1,4,8,6].

We Do · Practise with guidance

Insert key 5 into the sorted prefix [2,9] of [2,9,5,7].

Predict first

Which value must move first?

Model solution

9 moves one index right; 2 stays. Insert 5 at index 1 → [2,5,9,7].

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Question 1

Insert key 3 into [2,6,3,8], assuming [2,6] is sorted.

Question 2

Trace insertion sort fully on [5,1,4].

Question 3

Explain why insertion sort is efficient on almost-sorted input.

Exit Ticket · Answer before checking

Insertion sort always swaps each element with its neighbour.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. [2,4,5].

Q2. 3.

Q3. LIFO: last in, first out.

Q4. Track whether any swap occurred; if none occurred during a complete pass, stop.

You Do pages · Independent questions

Q1. Shift 6, then insert 3 → [2,3,6,8].

Q2. After key 1 → [1,5,4]; after key 4 → [1,4,5].

Q3. Few values need shifting, so its best-case time is O(n), though worst case is $O(n^2)$.

Exit Ticket · Model answer

False. It commonly shifts larger values and places the key in the gap.

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed: